LLMs Can Play (Global) Games

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Abstract

The communication channel that enables coordination is also the channel through which regimes suppress it. I embed seven large language models in the Morris–Shin (2003) regime change global game, conveying private signals as natural-language intelligence briefings rather than numeric values. Across 1,800 country–periods, join rates track the theoretical attack mass ($\bar{r} = +0.80$, $p < 0.001$ for every model), and a cost/benefit narrative sweep shifts equilibrium cutoffs in lockstep with theory ($r = 0.997$)—a 50 percentage-point swing that no text classifier can predict. I then exploit this behavioral regularity as a platform for information design. Censorship, ambiguity manipulation, and public signal injection each shift coordination outcomes in the directions predicted by Bayesian persuasion theory. The central finding concerns surveillance: when communications are monitored, expressed joining falls by -9.7 pp across five architectures—yet second-order beliefs shift by less than 1 pp. Agents self-censor while interpreting others' silence as genuine, sustaining the suppressed equilibrium because common knowledge of shared discontent never forms.

JEL: C72, C92, D82, D83, P16
Keywords: global games, regime change, LLM agents, information design, Bayesian persuasion, belief-action wedge, preference falsification

1 Introduction

The information channel is a trap. Coordination under uncertainty requires communication—agents must learn whether others share their willingness to act—but any channel that transmits this information is also the channel through which regimes suppress it. This paper documents this mechanism in the canonical setting: the global game of regime change (Morris and Shin, 2003), where private information resolves equilibrium multiplicity (Carlsson and van Damme, 1993; Frankel et al., 2003) and the stakes of coordination failure include bank runs (Diamond and Dybvig, 1983), currency attacks (Obstfeld, 1996), and political upheaval (Angeletos et al., 2007).

Global game models are highly tractable theoretically but notoriously difficult to test. Laboratory experiments must simplify the signal structure to numeric values on a screen, abstracting away the rich, multi-source information that characterizes real coordination problems (Heinemann et al., 2004, 2009; Szkup and Trevino, 2020). Field data from actual crises confounds strategic behavior with institutional differences and unobserved information structures. Neither setting can manipulate the information environment—the signal precision, correlation structure, and disclosure rules that are the key policy levers in the theory.

I address this gap by embedding seven architecturally distinct large language models (LLMs) in the Morris and Shin (2003) regime change game. Each agent receives a private signal $x_i = \theta + \varepsilon_i$, translated into a natural-language intelligence briefing whose content shifts monotonically with signal strength across eight evidence domains. No explicit payoff table is provided—the stakes are embedded in the narrative, forcing agents to form beliefs from language. The design enables what neither lab experiments nor field data can: testing information design predictions under near-full model complexity, at scale.

Three contributions. First, across 1,800 country–periods and seven models, LLM agents implement threshold policies closely consistent with the unique Bayesian Nash Equilibrium. Join rates track the theoretical attack mass with mean $r = +0.80$ ($p < 0.001$ for every model), and an orthogonal cost/benefit sweep shifts equilibrium cutoffs in lockstep with theory ($r = 0.997$). Unlike human subjects, who consistently exhibit level-k cognitive hierarchy behavior in coordination experiments (Szkup and Trevino, 2020), LLM agents' behavior overwhelmingly selects BNE over L_1 and L_2 alternatives.

Second, this behavioral regularity converts the experiment into a testbed for information design (Kamenica and Gentzkow, 2011; Goldstein and Huang, 2016). I test whether manipulating the signal structure—precision, correlation, or disclosure rules—shifts coordination outcomes as Bayesian persuasion theory predicts. Stability framing, public signal injection, and upper/lower censorship (Kolotilin et al., 2022) each produce the directional shifts the theory requires.

Third, and most importantly, I show that the communi-

cation channel that enables coordination is exploitable by an authoritarian planner. Surveillance produces a *belief–action wedge*: expressed joining falls by −9.7 pp across five architectures while second-order beliefs shift by less than 1 pp. Agents self-censor while interpreting others' silence as genuine—the regime prevents common knowledge of shared discontent from forming without changing the beliefs themselves. This is Kuran's (1991) preference falsification, generated not by hidden preferences but by poisoned communication.

Related literature. The paper connects three strands. The global game literature—from the generic uniqueness result of Carlsson and van Damme (1993) through the regime change application of Morris and Shin (2003) and the dynamic extensions of Angeletos et al. (2007) and Chassang (2010)—provides the theoretical framework. The key insight of Morris and Shin (2002), that public signals are overweighted because they predict others' actions, underlies the information design results. Laboratory experiments by Heinemann et al. (2004) confirmed threshold behavior; Szkup and Trevino (2020) found level-k thinking rather than BNE, motivating the cognitive-hierarchy comparison.[1]

The Bayesian persuasion framework (Kamenica and Gentzkow, 2011), applied to regime change by Goldstein and Huang (2016) and to optimal censorship by Kolotilin et al. (2022), provides the information design theory. The political economy of authoritarian control—Edmond (2013) on propaganda, Kuran (1991) on preference falsification, Shadmehr and Bernhardt (2011) on collective action—motivates the surveillance experiments. Empirical evidence that censorship targets coordination (King et al., 2013), surveillance suppresses expression (Penney, 2016; Stoycheff, 2016), and social media enables protest (Enikolopov et al., 2020) provides the real-world stakes.

On the methodological side, Horton (2023) proposed LLMs as "homo silicus." Subsequent work found LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Petrov et al., 2025). Huang et al. (2024) and Carlini et al. (2025) document that alignment shifts risk preferences, motivating narrative rather than tabular payoffs. Critical reviews by Larooij and Törnberg (2025a,b) identify validation challenges that I address through cross-generator robustness (Appendix F.9), scramble/flip falsification, and out-of-sample replication.

The paper proceeds as follows. Section 2 presents the theoretical framework. Section 3 describes the experimental design. Part I (Sections 4–5) establishes that LLM agents implement equilibrium-consistent threshold policies and that pre-play communication creates an exploitable information channel. Part II (Sections 6–7.1) shows how an information designer—through censorship, surveillance, and propaganda—can reshape coordination outcomes.

[1] Concurrent work by Ruano and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

2 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0, 1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta \leq 0$ represent regimes so weak they fall without opposition; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta. \tag{1}$$

Payoffs depend on the citizen's action and the outcome (Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \tag{2}$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003)**.** *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma \Phi^{-1}(\theta^*) \tag{3}$$

and $\theta^* = B/(B + C)$.

The *attack mass* $A(\theta)$—the theoretical fraction of the population that joins at regime strength θ—is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \tag{4}$$

This is a decreasing function of θ: weaker regimes face larger uprisings.

We define $J(\theta)$ as the *empirical join fraction*—the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$.

An information designer controls the mapping $\pi : \Theta \to \Delta(\mathcal{S})$ from states to signal distributions, but cannot control agents' actions. In my implementation, π is a deterministic mapping from regime strength θ to the parameters of the briefing generator—a system that produces a natural-language intelligence briefing from a z-score derived from the agent's private signal. All stochasticity enters downstream through the LLM's decoding.

The briefing generator has three control parameters: clarity (the width of the Gaussian kernel mapping z-scores to text, where wider kernels produce more ambiguous briefings), directional precision (the slope of the mapping from z-score to briefing sentiment, where steeper slopes produce more accurate signal reflection), and dissent framing (the floor on the probability that the briefing includes language about public discontent).

The designer concentrates manipulation near θ^* using a Gaussian proximity weight (Eq. 5):

$$w(\theta) = \exp\left(-\left(\frac{\theta - \theta^*}{h}\right)^2\right) \tag{5}$$

where $h = 0.15$ is the baseline bandwidth parameter.

The framework generates testable predictions for both the baseline game and information design. Table A1 in Appendix A reports the eight pre-specified hypotheses and their test results.

3 Experimental Design

The experiment has three parts. Part I tests whether LLM agents implement threshold-like policies in the global game when private signals are conveyed in natural language: a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. Part II takes the behavioral foundation as given and studies information design: stability/instability designs, censorship, public signal injection, and single-channel decomposition. Part III tests whether an authoritarian regime can exploit the communication channel through surveillance, propaganda, and their interaction. All LLM interactions use the same prompt structure across models.

A note on the state variable. In the theory (Section 2), θ is an unbounded fundamental with special roles for $\theta \leq 0$ (regime falls without opposition) and $\theta \geq 1$ (regime survives even unanimous attack). In Part I experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C). To evaluate monotonicity on a common scale, I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ (so $\theta^* = 0.50$) and $\sigma = 0.3$. In Part II, I keep $B = C = 1$ and restrict attention to a fixed θ-grid in $[0.20, 0.80]$ for comparability with the canonical $[0, 1]$ formulation. Table 1 summarizes the three regimes.

Table 1: Experiment regimes across paper sections.

Part	θ distribution	Baseline	Estimand
I (Alignment)	$\theta \sim \mathcal{N}(\bar{z}, 1)$	Pure game	Mean over θ draws
II (Info Design)	Fixed 9-point grid $[0.2, 0.8]$	Calibrated baseline	Mean over grid
III (Regime Tools)	Same as Part I	Comm. game	Mean over θ draws

Throughout, $\theta^* = B/(B + C)$ is the theoretical cutoff, $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$ the signal cutoff, and $\hat{\theta}^*$ the estimated cutoff from the fitted logistic $P(\text{join} \mid \theta) = 1/(1 + e^{-(\beta_0 + \beta_1\theta)})$. We write $\beta \equiv \beta_1$ for the logistic slope coefficient in tables.

For each country–period, nature draws $\theta \sim \mathcal{N}(\bar{z}, 1)$, where $\bar{z}$ is a public prior mean drawn randomly for each country. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$ and computes a z-score $z_i = (x_i - \bar{z})/\sigma$. Because agents observe only their private briefing and never the prior distribution or its parameters, the diffuse-prior equilibrium formula (Proposition 1) serves as the theoretical benchmark. The z-score is then translated into a multi-paragraph intelligence briefing by a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. Figure 1 summarizes the signal-to-text-to-decision pipeline.

Because the briefing generator maps z-scores monotonically to narrative, any sentiment-extracting model will produce a positive correlation with $A(\theta)$. The empirical contribution lies in the *quantitative structure*: sigmoid shape, payoff-narrative sensitivity, and robustness across seven models. Within-briefing falsification tests (Appendix F.6) confirm that the signal is distributed across all eight evidence domains.

Calibration adjusts a single parameter—the cutoff center—via a damped iterative procedure. The sigmoid shape is emergent and never optimized. Holdout validation confirms no overfitting; six uncalibrated models produce $|r| > 0.69$ (Appendix Table A6). No explicit payoff table is provided—the stakes are conveyed through narrative, since explicit payoffs caused models to ignore briefing content.

Part I has four treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the θ-to-signal link), and *flip* (negated z-scores). Part II implements information designs: *stability* (increased ambiguity near θ^*), *instability* (sharpened signals), *public signal injection*, and *upper/lower censorship* (Kolotilin et al., 2022). Part III tests authoritarian instruments: *surveillance* (monitoring warning in communication prompt) and *propaganda* ($m = 2, 5, 10$ regime plants).

I test seven models spanning six families (30B–235B parameters; Table 2). All experiments use $N = 25$ agents per country–period and $\sigma = 0.3$. Part I benchmarks use $B = C = 1$ ($\theta^* = 0.50$); payoff comparative statics vary narrative stakes. All LLM calls use temperature = 0.7 with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix G). Throughout, N without subscript refers to country–periods unless otherwise stated.

The unit of randomization is the country–period (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the country–period level to account for within-period correlation among agents sharing the same θ. For period-level correlations, I

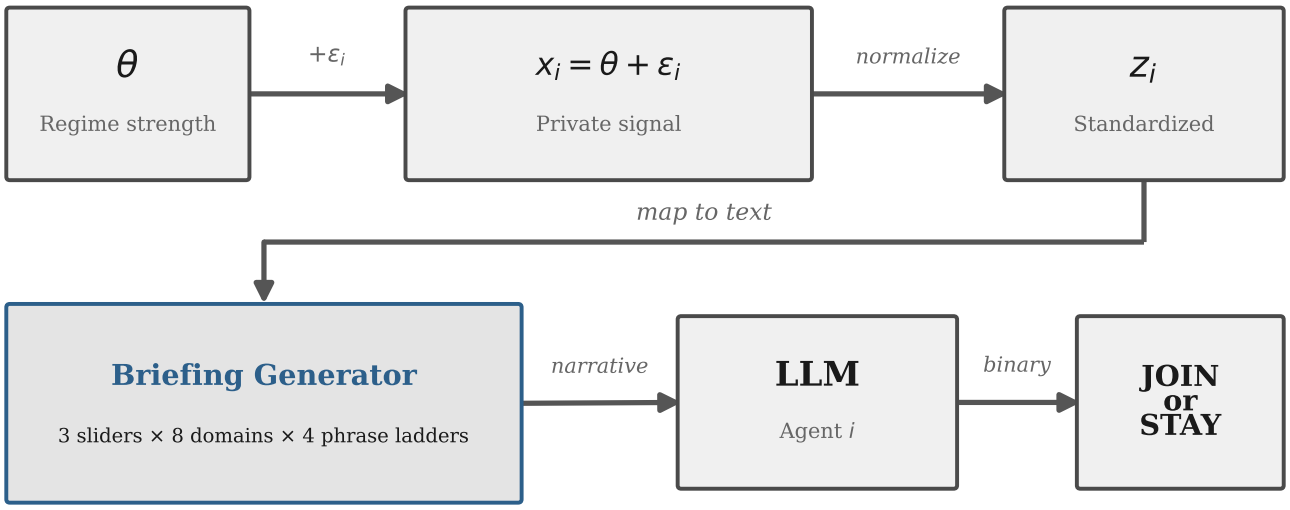


Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i, converted to z-scores and rendered into natural-language briefings via 8 evidence domains and 3 latent sliders. The briefing layer is deterministic; all stochasticity enters through LLM decoding.

Table 2: Model summary. Columns report country-period counts in the pure, communication, and falsification (scramble+flip) suites. All runs use $N = 25$ agents per period and $\sigma = 0.3$.

Model	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	Mistral	1000	1000	200
Llama 3.3 70B	Llama	100	100	200
Qwen3 30B	Qwen (MoE)	100	100	200
GPT-OSS 120B	GPT	200	200	1000
Qwen3 235B	Qwen (MoE)	200	200	200
Trinity Large	Arcee	100	100	200
MiniMax M2-Her	MiniMax	100	100	200
Total		**1800**	**1800**	**2200**

Notes: 7 models spanning 6 architecture families. N = country-periods with 25 agents each. All runs use $\sigma = 0.3$ and temperature = 0.7 unless otherwise stated.

report Fisher-z confidence intervals. The 25 agents within a period share the same θ and calibration; their decisions are conditionally independent given their private signals.[2]

For the information design experiments, I fix $B = C = 1$ (so $\theta^* = 0.50$) and a 9-point θ-grid in $[0.20, 0.80]$, with 30–60 repetitions per (design, θ) cell and 25 agents each. The primary model is Mistral Small Creative; cross-model replication uses five additional models.

Table 3 maps each treatment to the theoretical channel it tests, the directional prediction, and the observed result.

Of the eight pre-specified hypotheses, four survive Bonferroni correction at $\alpha = 0.05/8$: alignment (H1), flip (H3), stability (H5), and surveillance (H7). H2 (scramble) and H4 (communication) confirm null predictions by design. H6 (censorship) is significant at conventional levels ($p = 0.023$) but not after correction; H8 (propaganda) is directionally correct but non-significant. Exploratory analyses are reported with uncorrected p-values.

[2] I do not claim literal conditional independence—shared prompt structure and model weights introduce common factors. The clustering accounts for within-period correlation.

4 Results

Result 1 (Threshold-Policy Alignment)**.** *Across seven models and 1,800 country–periods in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$ (Eq. 4) averages $r = +0.80$ ($p < 0.001$ for every model).*

Figure 2 shows the empirical sigmoid for the primary model. Correlations range from $r = +0.75$ (Mistral) to $r = +0.84$ (Trinity), with the pooled OLS: $J = 0.13 + 0.54\,A(\theta)$ ($R^2 = 0.48$). The slope reflects classical measurement error attenuation from narrative-to-belief noise. All seven per-model correlations remain significant at $p < 0.001$ under country-clustered inference.

Alignment is stable across architectures (Figure 3): correlations span $r \in [+0.75, +0.84]$ despite parameter counts ranging from 30B to 235B. Mean join rates vary, reflecting model-specific action biases that shift the intercept but not the correlation. Estimated cutoffs vary ($\hat{\theta}^* \in [-0.32, +0.10]$); the disciplined evidence on cutoff location comes from comparative statics, where changing narrative stakes shifts $\hat{\theta}^*$ in lockstep with payoff theory

Table 3: Treatment map. Each row describes a treatment, the channel it tests, the unit of observation, the theoretical prediction, and the observed result. Part I tests whether LLM agents play the global game; Part II studies information design; Part III tests authoritarian exploitation of the communication channel. r denotes Pearson correlation; Δ is the change in mean join rate relative to the relevant baseline (percentage points).

Treatment	Part	Channel tested	Unit	Prediction	Observed
Core treatments					
Pure global game	I	Baseline monotonicity	Period	$r(J, A(\theta)) > 0$	$r = +0.80$
Communication	I	Pre-play messaging	Period	Ambiguous	+1.23 pp ($p = 0.327$)
Scramble (cross-agent)	I	Content vs. format	Period	$r \to 0$	$r = +0.05$
Flip	I	Signal direction	Period	Sign reversal	$r = -0.80$
B/C narratives	I	Payoff comparative statics	Period	Cutoff shifts	As predicted
Information design (Part II, primary model)					
Stability	II	Ambiguity near θ^*	Period	$\downarrow$ join	−9.0 pp
Instability	II	Sharpened signals	Period	$\downarrow$ join (clearer sorting)	−34.2 pp
Public signal	II	Common knowledge channel	Period	$\downarrow$ join (dominates private)	−39.2 pp
Censor (upper)	II	Pooling weak states	Period	Plateau below θ^*	−3.1 pp
Censor (lower)	II	Pooling strong states	Period	Plateau above θ^*	−1.9 pp
Authoritarian instruments (Part III)					
Surveillance	III	Preference falsification	Period	$\downarrow$ join	−13.5 pp
Propaganda (m=10)	III	Information contamination	Period	$\downarrow$ join, saturating	−2.3 pp (behavioral)
Within-briefing falsification					
Observation shuffle	I	Bullet ordering vs. content	Period	r unchanged	$r = +0.912$
Domain scramble (coord.)	I	Coordination-relevant domains	Period	$\lvert r\rvert$ falls if domains drive signal	$r = +0.921$
Domain scramble (state)	I	State-capacity domains	Period	$\lvert r\rvert$ falls if domains drive signal	$r = +0.928$

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass. Δ: change vs. relevant baseline (pp). Part I results are pooled across 7 models; Parts II–III use the primary model (Mistral Small Creative).

Table 4: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher-z confidence intervals in brackets for main treatments.

	Main treatments		Falsification			
Model	Pure	Comm	Scramble	Flip	n_{pure}	Mean join
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	+0.09	−0.81	1000	0.387
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02	−0.84	100	0.439
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13	−0.87	100	0.499
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01	−0.80	200	0.407
Qwen3 235B	+0.80 [0.74, 0.84]	+0.76 [0.70, 0.81]	+0.06	−0.85	200	0.416
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05	−0.80	100	0.464
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01	−0.62	100	0.436
Pooled	+0.76 [0.74, 0.78]	+0.75 [0.73, 0.77]	+0.03	−0.79	1800	0.409
Mean across models	+0.80	+0.78	+0.05	−0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$. 95% Fisher-z confidence intervals in brackets. Join fraction uses valid decisions only (excluding parse errors). Pooled: all country-periods concatenated; Mean: equal-weighted average across models.

(Table 5, Figure 4). A structural estimation against cognitive hierarchy benchmarks confirms these threshold policies are remarkably sophisticated: LLM behavior closely fits the unique Bayesian Nash Equilibrium (BNE), rejecting the level-k behavior typically observed in human laboratory subjects.

Human subjects in coordination game experiments consistently exhibit level-k cognitive hierarchy behavior rather than full equilibrium reasoning (Szkup and Trevino, 2020). To test whether LLM agents play the BNE or a level-k heuristic, I evaluate theoretical attack masses under BNE, L_1 (agents best-respond believing others randomize uniformly), and L_2 (agents best-respond believing others play L_1). While the thresholds are identical when $\theta^* = 0.50$, the cost/benefit sweep uniquely separates them. Across the seven cost/benefit conditions, the empirical join fraction overwhelmingly selects BNE (RMSE = 0.199, $r = +0.737$) over L_1 (RMSE = 0.307, $r = +0.445$) and L_2 (RMSE = 0.348, $r = +0.354$). The agents' threshold policies approximate the full Bayes-Nash Equilibrium.

The claim is behavioral: agents do not observe (B, C), σ, or N, yet the *pattern* of responses—monotone sigmoid,

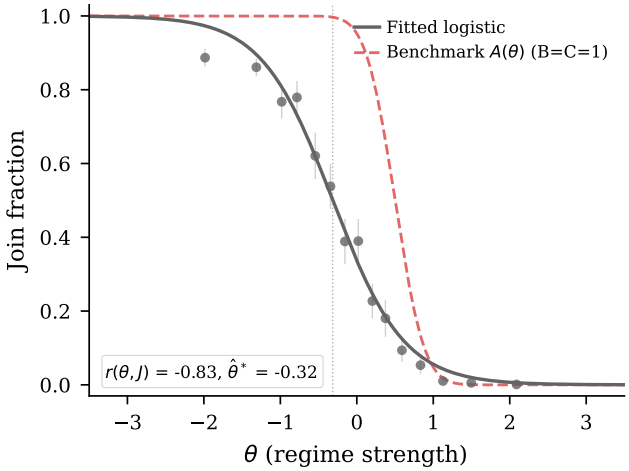


Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 periods). The empirical sigmoid is shifted leftward ($\hat{\theta}^* = -0.33$) relative to theory ($\theta^* = 0.50$). Cross-model results in Table 4 (mean $r = +0.80$, all $p < 0.001$).

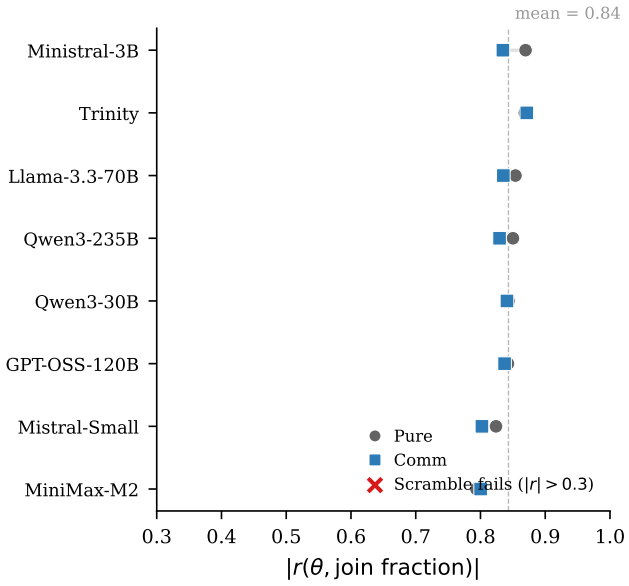


Figure 3: Cross-model signal monotonicity. Points report $r(J, A(\theta))$ under pure and communication.

payoff-sensitive cutoff, belief-mediated decisions—is robust across architectures and falsifiable. Three identification strategies establish that the regularity reflects more than text classification.

Cost/benefit identification. Varying narrative stakes while holding briefings fixed shifts cutoffs in lockstep with payoff theory. High-cost narratives drop mean joining to 19.0% ($\hat{\theta}^* = 0.132$); low-cost raises it to 69.3% ($\hat{\theta}^* = 0.716$)—a 50 pp swing invisible to any text classifier. A sweep across seven B/C ratios confirms $\hat{\theta}^*$ tracks θ^* with $r = 0.997$ ($p < 0.001$; Figure 4).

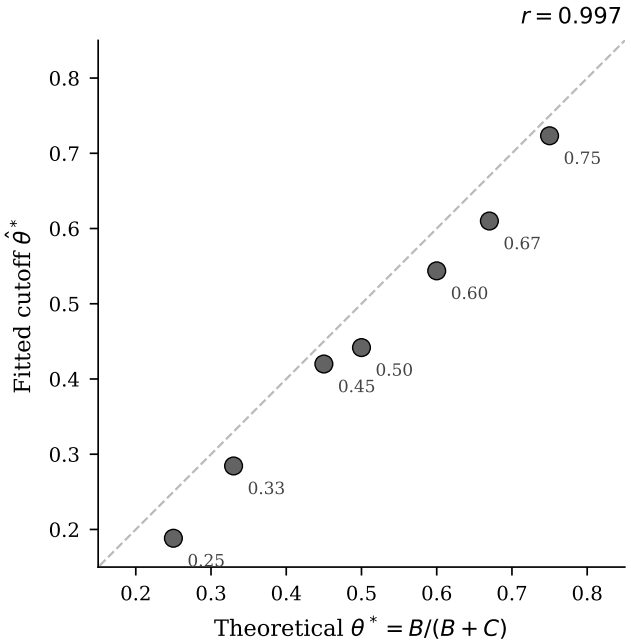


Figure 4: Fitted cutoff $\hat{\theta}^*$ vs. theoretical θ^* across seven B/C ratios ($r = 0.997$).

Belief mediation. Stated probability judgments mediate the signal-to-action pathway (Pseudo $R^2 = 0.975$; Table A15). Beliefs track the theoretical success probability ($r = +0.78$) and predict actions beyond what signals alone explain (partial $r = +0.64$). Pre-decision elicitation ($r_{\text{pre}} = +0.82$) rules out ex-post rationalization.

Classifier baselines. Text features alone cannot reproduce LLM behavior. A bag-of-words classifier achieves 88.4% accuracy in-distribution but fails to predict the surveillance chilling effect (+13.0 pp gap; Table A3 in Appendix C). Additional identification tests—coordination-cue manipulation and construct validity—are reported in Appendix C.

The correlation is invariant to decoding temperature ($r \in [+0.76, +0.79]$ across $T \in \{0.3, 0.7, 1.0\}$; Appendix F.5). What matters for Part II is that the behavioral regularity is robust enough to serve as a platform for studying how information structures shift coordination outcomes.

Result 2 (Signal Dependence)**.** *Cross-period scrambling*

Table 5: Cost/benefit narrative comparative statics. High cost: narrative emphasizes severe reprisals for failed action. Low cost: narrative emphasizes minimal consequences. Theory predicts higher perceived cost lowers the cutoff (less joining).

Design	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$ (SE)	Δ (pp)
Baseline	270	0.409	+0.88	0.39 (0.007)	—
High cost	270	0.190	+0.87	0.13 (0.010)	-21.9
Low cost	270	0.693	+0.88	0.72 (0.007)	+28.4

Notes: Primary model, information design θ-grid. Identical briefings across conditions; only the benefit/cost narrative framing varies. Δ: change vs. baseline mean join (pp).

of briefings reduces the mean within-country correlation from $r = +0.80$ to $r = +0.05$ across seven models. The pooled correlation drops from $r = +0.76$ to $r = +0.03$ (Fisher $z = 25.09$, $p < 0.001$).

The scramble preserves the marginal distribution of briefing content but breaks the mapping from each period's θ to the signals agents receive.[3]

Result 3 (Signal Direction)**.** *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across seven models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure → scramble → flip pattern replicates across all seven models (Figure 5; Table 4).

A text-only baseline ($\hat{p}_{\text{join}} = 1 - d$, where d is the briefing's internal direction score) correlates with LLM decisions at $r = 0.80$, but the LLM's join curve is substantially steeper (Figure 6), producing threshold-like behavior rather than linearly tracking tone.

5 Communication

Result 4 (Communication opens a channel without shifting coordination)**.** *Pre-play communication changes the mean join rate by -0.4 pp, from 0.435to 0.431, averaged across seven models. In the pooled sample, the unpaired difference is +1.23 pp ($p = 0.327$); effects vary in sign across models and are concentrated in weak-regime environments.*[4]

Communication preserves the signal structure ($r = +0.78$ vs. $+0.80$ under pure; Figure 7). The effect on join rates is heterogeneous across models and near zero on average, consistent with passive Bayesian updating concentrated in weak-regime environments.

How informative are the messages? Regressing θ on message features yields $R^2 = 0.12$—messages carry real information, noisily. This informativeness is what makes the channel exploitable: surveillance degrades R^2 to 0.02 (Section 7), an 80% reduction. Communication creates the information topology that authoritarian instruments exploit. With group-size knowledge, the communication premium reverses (-3.4 pp; Appendix F).

Table 6: Information design treatment summary (primary model: Mistral Small Creative). r is the Pearson correlation $r(J, A(\theta))$ between join fraction and theoretical attack mass.

Design	Mean	r	Δ (pp)	N
Baseline	0.409	+0.88	—	270
Stability	0.319	+0.62	-9.0	270
Instability	0.067	+0.74	-34.2	540
Public signal	0.017	+0.53	-39.2	540
Scramble	0.121	−0.03	-28.8	270
Flip	0.663	−0.82	+25.5	270

Notes: Data from the primary model (pure treatment; $\theta \in [0.20, 0.80]$ on a 9-point grid; N=25 agents per period). Mean join uses valid decisions; r is Pearson $r(J, A(\theta))$ across rep-level periods.

6 Information Design

Each information design modifies the briefing generator's slider functions near θ^* while agents' decision process remains unchanged, testing whether theoretical predictions (Goldstein and Huang, 2016; Kolotilin et al., 2022) produce the predicted shifts when agents are LLMs.

Figure 8 displays the design effects; Table 6 summarizes the main results. The baseline condition produces a mean join rate of 40.9%($r = +0.885$, $p < 0.001$).

Result 5 (Information Design Shifts Coordination)**.** *All information design families produce measurable shifts in coordination relative to baseline.*

The stability design suppresses coordination: mean join falls from 40.9%to 31.9%(-9.0 pp), and the regime-fall rate drops from 43.3% to 34.1%. The instability design paradoxically reduces mean joining to 6.7%(-34.2 pp) because sharper signals suppress joining in strong states more than they boost it in weak states, given the left-shifted baseline sigmoid. The public signal produces the largest reduction (1.7%, -39.2 pp) via the strategic complementarity of Morris and Shin (2002): agents overweight public signals because such signals predict others' beliefs, crowding out private information. Treatment effects by θ are shown in Figure A4 (Appendix D).

Kolotilin et al. (2022) proved that the optimal information structure is upper censorship—one-sided pooling

[3]Because the cross-period permutation operates within countries, all scramble correlations use within-country (country-demeaned) Pearson r.

[4]I report the unpaired (between-period) test as the primary specification because pure and communication treatments use independent θ draws, so country–periods are not naturally matched. A paired test that matches periods within each model by θ-rank yields a significant positive effect (+5.5 pp, $p < 0.001$, $n = 680$ pairs), reflecting within-θ variation that the unpaired test averages over. The qualitative conclusion—that the effect is small relative to baseline variation and heterogeneous across models—is robust to both approaches.

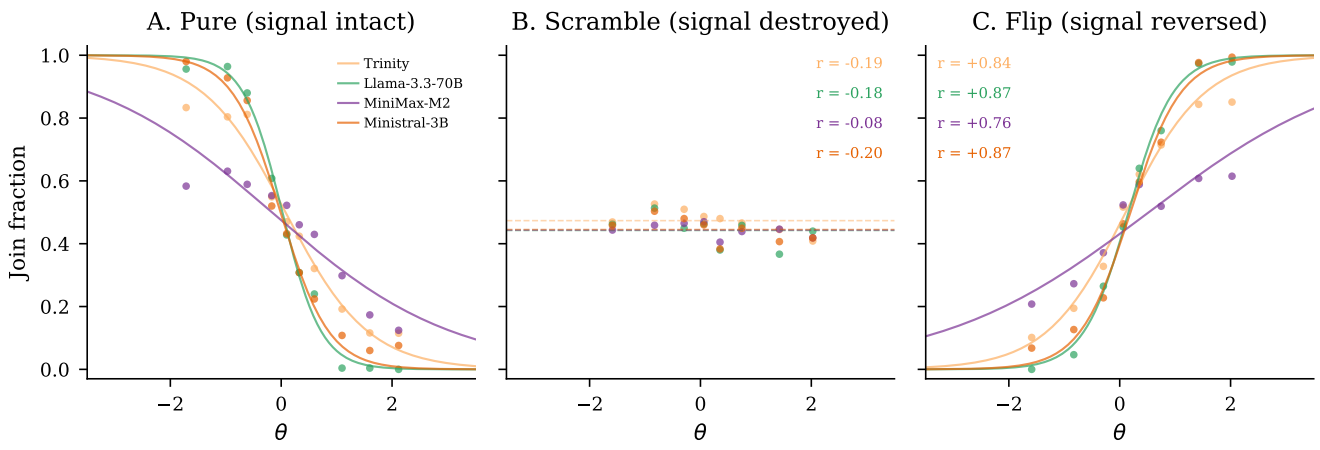


Figure 5: Falsification triptych. *Left:* Pure (mean $r = +0.80$). *Center:* Scramble ($r = +0.05$). *Right:* Flip ($r = -0.80$).

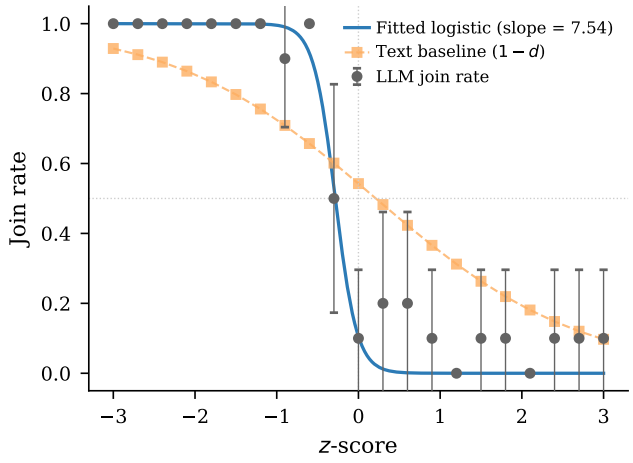


Figure 6: Text baseline test. The LLM join curve is steeper than the naïve text predictor $(1 - d)$, indicating processing beyond surface sentiment.

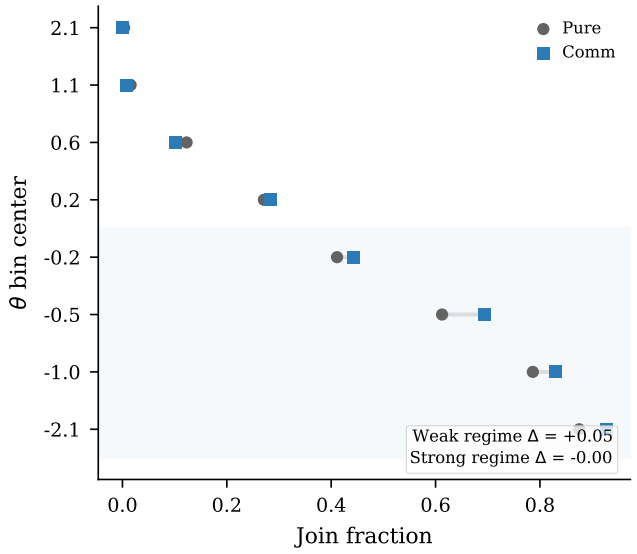


Figure 7: Communication effect by regime strength, pooled across seven models.

that conceals unfavorable states. Upper censorship pools weak-regime states ($\theta \leq \theta^*$) into a neutral signal; lower censorship pools strong-regime states.

Result 6 (Upper Censorship Suppresses Joining in Weak States)**.** *Upper censorship lowers the mean join rate to 37.7%(-3.1 pp vs. baseline) and attenuates the action–attack-mass correlation ($r = +0.722$). The effect is concentrated in the censored region ($\theta \leq \theta^*$), where weak-regime states are pooled to a neutral briefing and join rates flatten.*

Pooling generates a flat join-rate "plateau" in the censored region (Figure 9): when agents cannot distinguish $\theta = 0.20$ from $\theta = 0.50$, they behave as if the regime is borderline rather than clearly weak.

A common-knowledge test makes the censorship rule transparent. When agents are told "regime censors are suppressing unfavorable intelligence above a certain severity threshold," the mean join rate returns close to the uncensored baseline (43.2% vs. 40.9% baseline; Table A11)—common knowledge of the censorship rule largely neutralizes the pooling distortion, consistent with theory assuming informed receivers (Kolotilin et al., 2022). The naïve censorship result extends the theory to receivers who fail to account for the information structure—a realistic scenario where the censorship apparatus is opaque.

Result 7 (Lower Censorship Reverses Comparative Statics)**.** *Lower censorship produces a mean join rate of 39.0%(-1.9 pp vs. baseline) and flips the comparative statics: the action–attack-mass correlation reverses sign ($r = -0.750$). Below θ^*, censoring favorable signals suppresses joining (agents see only weak-regime cues); above θ^*, pooling strong-regime states to a neutral briefing raises join rates sharply—agents who would otherwise see dis-*

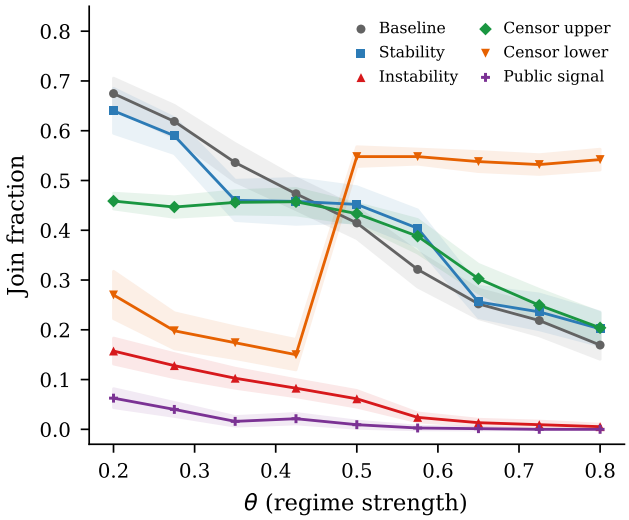


Figure 8: Join fraction vs. θ under six information designs (Mistral Small Creative).

couraging intelligence now receive uninformative briefings and default toward joining. The discontinuity at θ^* is consistent across repetitions (within-cell $\sigma < 0.04$) and replicates across models, though the direction of the reversal is model-dependent (Appendix F).

Falsification within information design confirms signal dependence: scrambling collapses the correlation to $r = -0.033$ ($p = 0.59$); flipping inverts it to $r = -0.820$ ($p < 0.001$). A harder scramble generating all briefings from $\theta = 0.5$ produces $r \approx 0$ for both Mistral and Llama (Figure A5 and bandwidth sensitivity in Appendix D).

7 Surveillance

Kuran (1991) labeled the mechanism underlying authoritarian stability "preference falsification"; I adopt the more operationally precise term *belief–action wedge* because my experimental design measures the gap between stated beliefs and expressed actions rather than hidden preferences *per se*.

In the surveillance treatment, the communication prompt is augmented with a warning that communications are being monitored by regime security services. The manipulation affects only the communication phase; the decision prompt is unchanged. The isolation is architectural: each LLM call is stateless, and the decision-stage prompt contains no reference to surveillance. Any difference in join rates is most parsimoniously explained by agents self-censoring their communications, not by a direct change in the perceived cost of joining.

Two placebo variants—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline (+2.6 pp and +4.0 pp, both $p > 0.2$; Appendix F.6), confirming that the chilling effect is specific to the surveillance warning's strategic content.

Result 8 (Surveillance Produces a Large Chilling Effect)**.** *In the primary model (Mistral Small Creative), surveillance reduces mean join rates from 41.1% to 27.6%, a difference of -13.5 percentage points ($p < 0.001$). The action–attack-mass correlation is preserved under surveillance ($r = +0.68$ vs. $+0.74$ baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance contaminates the information environment with self-censored messages, pushing join rates well below the communication baseline (Figure 10). The effect replicates across five models: Mistral (−13.5 pp), GPT-OSS (−11.9 pp), Qwen3 30B (−10.9 pp), Llama (−8.9 pp), and Qwen3 235B (−3.3 pp), averaging −9.7 pp.

Indirect channel: self-censored messages degrade information. Surveillance shifts stated beliefs by only −0.7 pp ($p = 0.18$, not significant) while shifting join rates by −12.9 pp ($p < 0.001$). A decomposition formalizes this (Eq. 6):

$$\underbrace{\Delta\bar{J}}_{\text{total}} = \underbrace{\frac{\partial \bar{J}}{\partial \bar{b}}\,\Delta\bar{b}}_{\text{belief updating}} + \underbrace{\Delta\bar{J}\big|_{\bar{b}\text{ fixed}}}_{\text{belief–action wedge}} \tag{6}$$

Virtually the entire effect passes through the belief–action wedge—agents change what they *do* without changing what they *believe*. Elicited punishment risk is approximately 8.0/10 across all conditions with negligible differences between JOIN and STAY agents (Table A19).

Second-order beliefs are unchanged by surveillance (31.0% → 30.9%, $p = 0.5866$), revealing the *strategic update gap*: each agent self-censors while interpreting others' silence as genuine, sustaining the suppressed equilibrium because common knowledge of the chilling effect never forms. The R^2 of message content on true θ drops from 0.12 to 0.02—an 80% reduction in informational content.

Direct channel: decision-stage suppression. A hold-fixed test replays baseline messages verbatim while warning that actions may be monitored. Join rates drop by −20.9 pp ($p < 0.001$), confirming that surveillance can also operate through a direct decision-stage channel. This channel is absent in the main design (where the decision prompt contains no surveillance reference); the main treatment isolates the indirect channel.

Message content confirms self-censorship: "act" drops from 19.0% to 3.8% of messages under surveillance; "collapse" from 11.6% to 0.6%. Among agents who privately decide to JOIN, only 19.0% send action-signaling messages under surveillance vs. 46.1% under regular communication.

Instrument interactions. When propaganda ($m = 5$) and surveillance are combined, the mean join rate among

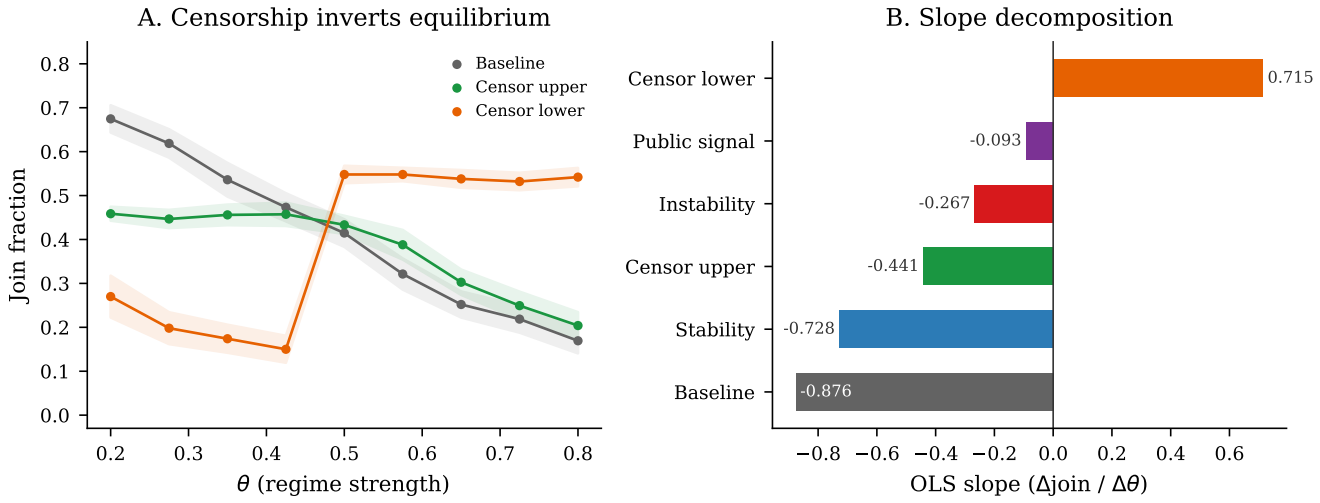


Figure 9: Censorship effects. *Left:* Upper censorship creates a flat plateau; lower censorship produces a sharp reversal. *Right:* OLS slope decomposition.

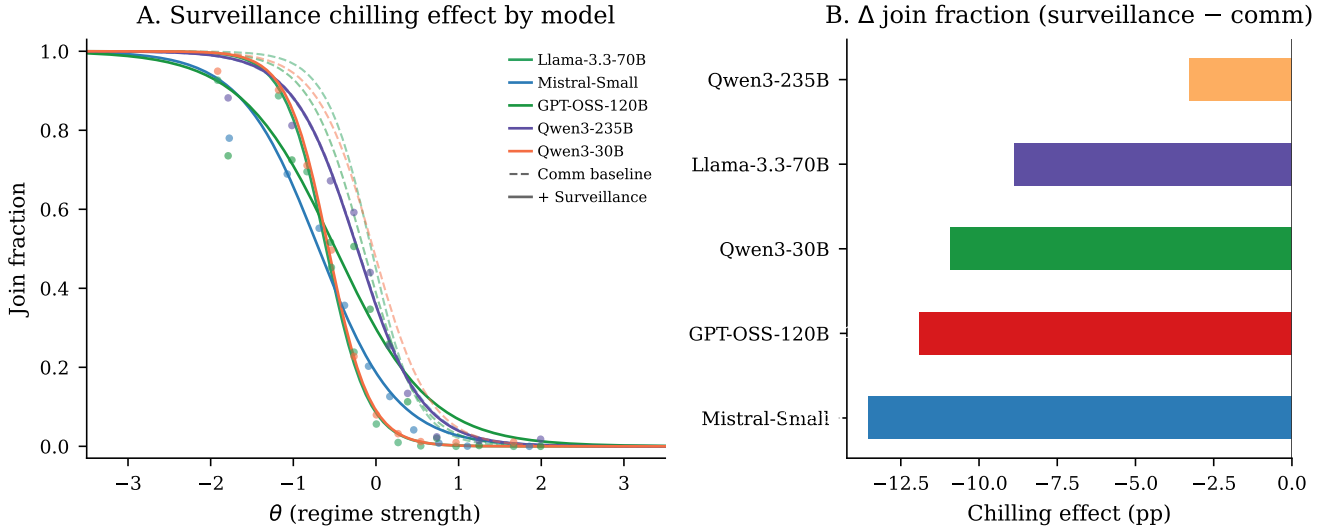


Figure 10: Surveillance reduces join rates across five models, averaging −9.7 pp.

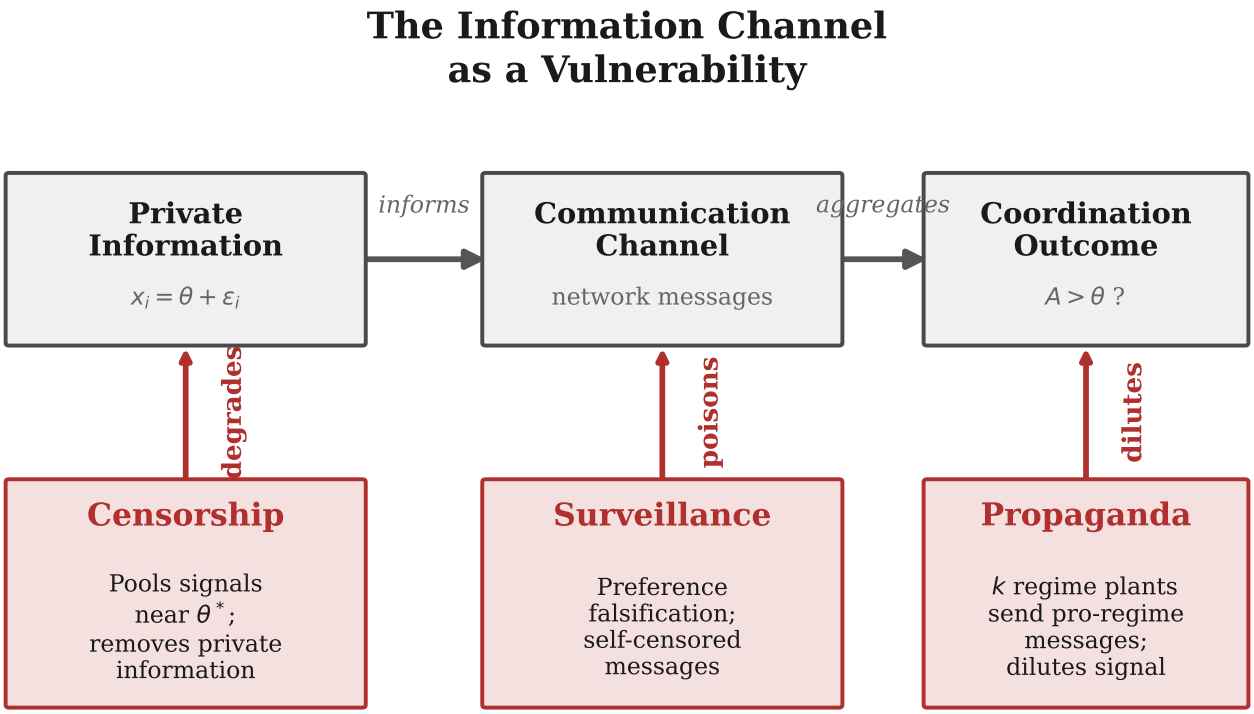
real citizens falls to 24.3%, a reduction of −16.8 pp from the communication baseline (41.1%). The sum of individual effects is −15.5 pp (surveillance −13.5 pp + propaganda −2.0 pp), so the combined effect (−16.8 pp) is approximately additive: once surveillance has suppressed expressed dissent, propaganda adds only modest additional deterrence.

Surveillance and censorship, by contrast, interact super-additively (Table 7): the marginal effect of surveillance is substantially larger under censorship than at baseline. Censorship degrades private signal quality, making the communication channel relatively more important, and surveillance poisons that channel. With both instruments active, agents have neither private signals to trust nor authentic messages to learn from (Figure 11). The interaction is heterogeneous across architectures (Table 8): the joint effect depends on model-specific resolution of pooled private signals and self-censored messages.

Table 7: Surveillance × censorship interaction in the communication game (primary model: Mistral Small Creative).

Design	No Surv.	Surv.	Δ (pp)
Baseline	0.030	0.009	-2.1
Upper cens.	0.151	0.037	-11.4
Lower cens.	0.177	0.042	-13.5

Notes: "No Surv." uses the communication infodesign grid. "Surv." uses the same grid with surveillance active during messaging. All entries are means of `join_fraction_valid`.



Instrument Interactions

Surv. + Propaganda:

Approximately additive

Surv. + Censorship:

Super-additive: -11.4 pp under censorship vs -2.1 pp at baseline

The regime does not need to change what citizens believe; it needs only to make them uncertain about each other.

Figure 11: How authoritarian instruments attack the coordination chain. Surveillance and censorship are complements; propaganda saturates quickly.

Table 8: Cross-model surveillance × censorship interaction. All conditions run under surveillance with communication. Δ columns show the change relative to the surveilled baseline.

	Mean join (surv.)			Δ (pp)	
Model	Baseline	Upper cens.	Lower cens.	Δ upper	Δ lower
Mistral Small Creative	0.009	0.037	0.042	+2.8	+3.3
Llama 3.3 70B	0.114	0.039	0.115	-7.5	+0.1
GPT-OSS 120B	0.316	0.177	0.312	-13.9	-0.4
Qwen3 235B	0.336	0.321	0.468	-1.5	+13.1

Notes: Communication treatment, information design θ-grid, surveillance active during messaging. Δ: change vs. surveilled baseline (pp). $N = 25$ agents per period.

Table 9: Propaganda and surveillance effects (primary model: Mistral Small Creative). "All" includes propaganda agents; "Real" excludes them (computed from logs). Δ is the change in real-agent mean join vs. baseline communication.

	Mean join			
Treatment	All	Real	r	Δ (pp)
Comm (baseline)	0.411	0.411	+0.74	—
Prop $m = 2$	0.375	0.407	+0.72	-0.4
Prop $m = 5$	0.313	0.391	+0.72	-2.0
Prop $m = 10$	0.233	0.388	+0.72	-2.3
Surveillance	0.276	0.276	+0.68	-13.5
Prop+Surv	0.194	—	+0.61	—

Notes: Primary model (Mistral Small Creative), communication treatment. "All" includes propaganda plant agents; "Real" excludes them. Δ: change in real-agent mean join vs. baseline communication (pp).

7.1 Propaganda

Edmond (2013) modeled propaganda as the regime shifting citizens' signal distributions. I introduce m propaganda agents (regime plants, always STAY) into the 25-citizen communication network. The *overall* join rate includes plants; the *real citizen* rate isolates the behavioral effect (Figure 12).

Result 9 (Propaganda Suppresses Coordination Primarily Through Mechanical Dilution)**.** *Mean join fraction (including plants) falls from 41.1% ($m = 0$) to 37.5% ($m = 2$), 31.3% ($m = 5$), and 23.3% ($m = 10$). However, the behavioral effect on real citizens is much smaller and saturates: 41.1% ($m = 0$), 40.7% ($m = 2$, -0.4 pp), 39.1% ($m = 5$, -2.0 pp), 38.8% ($m = 10$, -2.3 pp).*

The mechanical channel (plants always STAY) is approximately linear in m; the behavioral channel saturates quickly—doubling plants from 5 to 10 produces no additional effect (-0.3 pp, $p = 0.6676$). The pattern replicates with Llama (-2.7 pp at $m = 5$).

Message content reveals the mechanism (Table 9). Propaganda injects regime-loyal vocabulary ("loyal" rises from 1.5% to 11.4% at $m = 10$) while coordination language declines ("ready" from 30.5% to 18.5%). Among real agents who STAY, caution-coded messages rise from 24.2% to 38.2%—agents echo propaganda. Among those who JOIN, action signaling remains stable at $\approx 86\%$.

8 Conclusion

The information channel is a trap, and this paper has documented the trap's mechanism. LLM agents implement threshold policies closely consistent with the unique Bayes-Nash Equilibrium, and this behavioral regularity converts the experiment into a testbed for information design. The key finding is that the communication channel enabling coordination is the same channel through which regimes suppress it: surveillance induces a belief–action wedge where agents self-censor while interpreting others' silence as genuine, preventing common knowledge of shared discontent from forming. Censorship and surveillance are complements—degrading private signals makes the communication channel more important, and surveillance poisons that channel.

Three limitations warrant discussion. First, whether the underlying mechanism is approximate Bayesian reasoning or a learned heuristic that happens to match BNE remains open. The behavioral claim—that the pattern is robust, replicable, and falsifiable—does not require resolving this question, but interpreting the welfare implications requires it. Second, the experiments are static; real coordination problems are dynamic (Angeletos et al., 2007; Chassang, 2010), and the dynamics of belief updating under persistent surveillance may differ from the one-shot effects documented here. Third, external validity remains the fundamental challenge for "homo silicus" (Horton, 2023): what LLM agents do in a coordination game is evidence about the structure of the game, not necessarily about how humans would play it.

Despite these limitations, the framework offers something neither laboratory experiments nor field data can: a setting where the information environment—signal precision, correlation structure, disclosure rules, communication monitoring—can be manipulated while holding everything else fixed. The experimental template is reusable, and the data and code are publicly available.

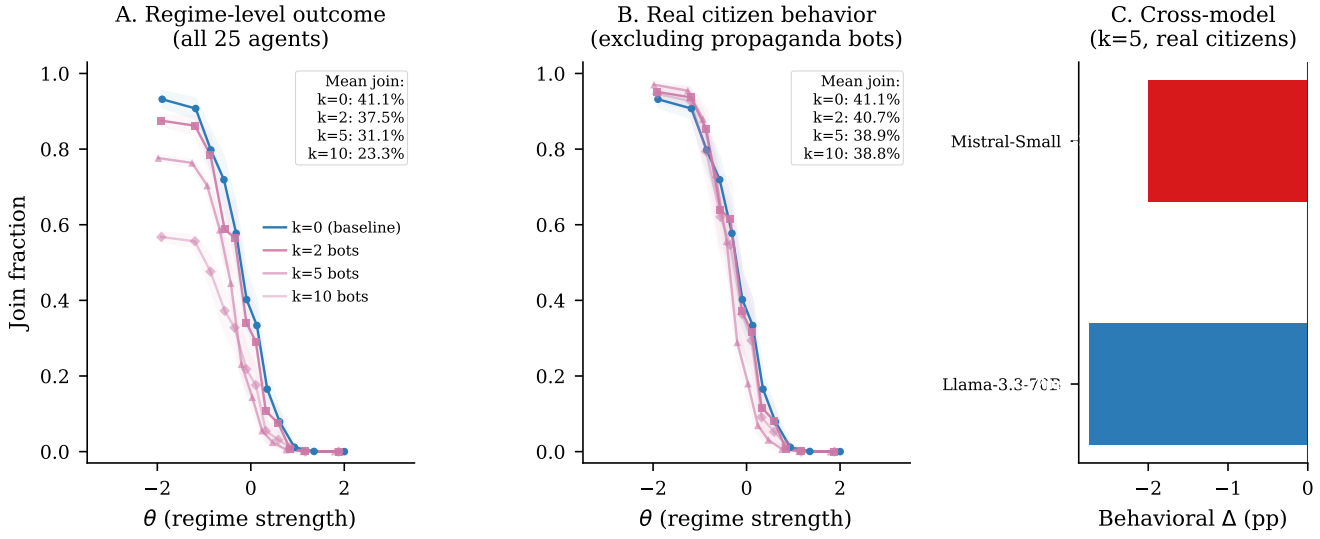


Figure 12: Propaganda dose-response: Mistral (primary) and Llama (replication).

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A Experimental Design Details

Table A1 reports the eight pre-specified hypotheses, their estimands, test statistics, and outcomes. H1–H4 are tested on pooled data across all seven models; H5–H8 use the primary model (Mistral Small Creative). Bonferroni-corrected threshold: $\alpha = 0.00625$.

B Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision ("How likely is it that the regime falls this period?"). Figure A1 documents calibration quality against the theoretical posterior; Figure A2 shows second-order beliefs (predictions about others' join rates) are unchanged by surveillance; Figure A3 confirms monotone signal–belief mapping nonparametrically.

C Identification Test Details

Table A2 reports OLS and logit estimates for the common-knowledge framing experiment. Table A3 compares text-classifier accuracy under baseline and surveillance conditions, establishing that the chilling effect is invisible to any classifier conditioned on the intelligence briefing alone. Table A4 repeats the classifier exercise across the cost/benefit sweep.

D Information Design Details

Figure A4 plots $\Delta(\theta)$ — the treatment-induced shift in join rate as a function of regime strength — for each design. Figure A5 confirms internal validity: the scramble and flip falsifications replicate within the information design θ-grid, ruling out confounds from the grid's θ range. The proximity bandwidth h controls how sharply manipulation concentrates around θ^*; the baseline $h = 0.15$ matches the θ-grid spacing and is approximately optimal (Table A9, Appendix F).

D.1 Hard Scramble

A harder scramble test generates *all* briefings from a single fixed state ($\theta = 0.5$), severing any possible correlation between θ and briefing content by construction. Under this hard scramble, $r(J, A(\theta)) = +0.05$ ($p = 0.42$) for Mistral and $r = +0.11$ ($p = 0.08$) for Llama—both indistinguishable from zero.

E Decomposition: Which Channel Drives the Stability Effect?

The stability design manipulates three channels simultaneously (direction, clarity, dissent). To determine which drives the effect, I run three single-channel treatments, each activating only one manipulation while holding the other two at baseline.

Each channel alone produces a modest suppression of joining relative to baseline (Figure A6): clarity only (-0.4 pp), direction only (-2.0 pp), and dissent only (-0.2 pp).

Summing the single-channel effects (Table A5) yields -2.6 pp, far smaller in magnitude than the bundled stability design effect (-9.0 pp). This implies strong super-additivity: each channel alone has a modest effect, but the combination produces a suppression more than three times the sum of its parts. The channels are complements—ambiguity (clarity), directional flattening, and dissent framing interact to undermine coordination.

F Robustness

Threshold-policy alignment is stable to agent count ($r \in [+0.60, +0.68]$ across $N \in \{5, 10, 25, 50, 100\}$), network density ($r = +0.66$ at $k = 8$ vs. $+0.65$ at $k = 4$), and mixed-model populations ($r = +0.77$ pure, $r = +0.75$ communication; Figure A7).

F.1 Robustness Checks

Without calibration, all six models show $r > 0.69$ and five exceed $r > 0.75$ (Table A6). Deliberately miscalibrating two models (± 0.3 shift) leaves r virtually unchanged ($r \in [+0.75, +0.76]$; Table A7): calibration adjusts the *level* but does not create the *slope*. Table A8 and Figure A8 report calibration quality across all seven models (r ranges from $+0.75$ to $+0.84$).

The proximity bandwidth h controls how sharply information design concentrates manipulation around θ^*. Treatment effects are qualitatively robust across $h \in \{0.05, 0.15, 0.30\}$ but magnitudes vary: the stability effect peaks at $h = 0.15$ because this bandwidth matches the θ-grid spacing, and the baseline $h = 0.15$ is approximately optimal for detecting treatment effects (Table A9).

F.2 Cross-Model Replication of Information Design

The flip inversion replicates across all five models (flip $r < -0.43$; Figure A9, Table A10). The within-country scramble shows heterogeneity: Mistral, GPT-OSS, and Qwen3 235B collapse cleanly ($r \approx 0$), but Llama retains $r = +0.81$ because the within-country permutation preserves ecological structure through the country prior mean. The hard scramble ($\theta = 0.5$ for all briefings) collapses Llama to $r = +0.11$ ($p = 0.08$), confirming residual ecological signal rather than a falsification failure.

Table A1: Pre-specified hypotheses and test results. H1–H4 use pooled Part I data across all seven models; H5–H8 use the primary model (Mistral Small Creative). "Supported" indicates whether the data pattern matches the hypothesis at $\alpha = 0.05$.

H	Hypothesis	Estimand	Null	Test	Stat	p	Supported?
H1	Sigmoid Response	$r(J, A(\theta))$	$r = 0$	Pearson	0.757	<0.001	Yes
H2	Scramble Falsification	r(scramble)	$r = 0$	Pearson	0.029	0.343	Yes
H3	Directional Sensitivity	r(flip)	$r \geq 0$	Pearson (1-sided)	-0.791	<0.001	Yes
H4	Communication Channel	Δ_{pp}	$= 0$	t-test	0.981	0.327	No (ambiguous)
H5	Ambiguity Pooling	Δ_{pp}	$= 0$	t-test	-5.322	<0.001	Yes
H6	Censorship Distortion	Δ_{pp}	$= 0$	t-test	-2.284	0.023	Yes
H7	Surveillance Chilling	Δ_{pp}	$= 0$	t-test	-8.277	<0.001	Yes
H8	Propaganda Dose-Response	Δ_{pp}	$= 0$	t-test	-0.892	0.372	No (ambiguous)

Notes: H1–H4: pooled across all seven models. H5–H8: primary model (Mistral Small Creative). Bonferroni-corrected threshold: $\alpha = 0.05/8 = 0.00625$.

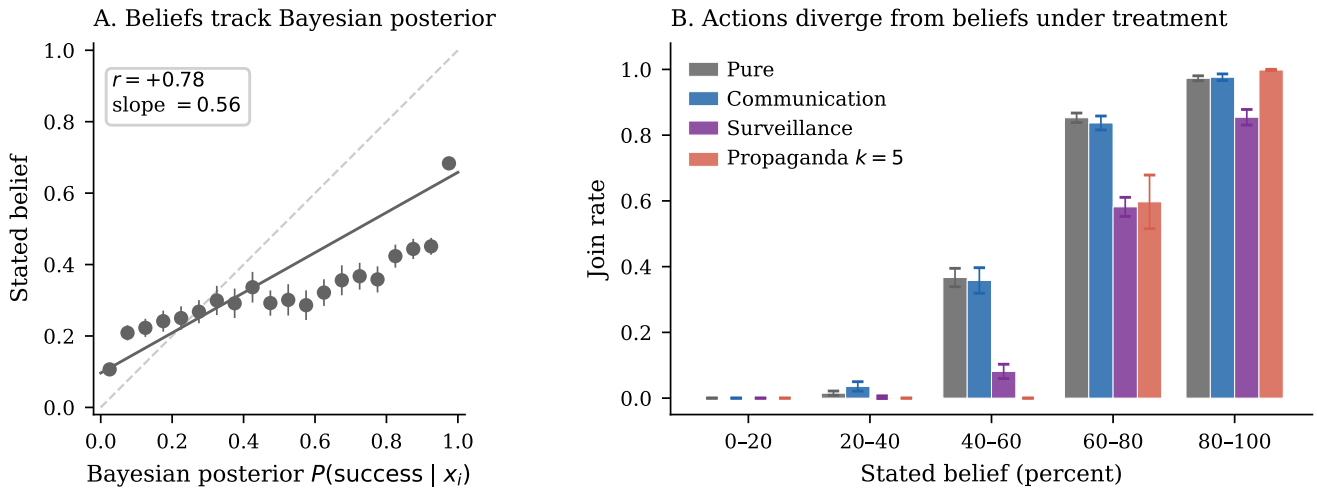


Figure A1: Belief elicitation (Mistral). *Left:* Stated beliefs vs. theoretical $P(\text{success} \mid x_i)$ ($r = +0.78$). *Right:* Join rate by belief bin.

Table A2: Common knowledge × coordination intensity. Each cell reports mean join rate (270 country–periods). The CK main effect is +6.1 pp ($p = 0.0003$); the interaction is -1.2 pp ($p = 0.62$).

	Low coord	High coord	Δ (coord)
Private framing	37.4%	37.8%	+0.4 pp
CK framing	43.5%	42.7%	-0.8 pp
Δ (CK)	+6.1 pp	+4.9 pp	

Notes: Primary model, information design θ-grid. 270 country-periods per cell. CK framing: agents told their briefing is "widely shared." High-coord: coordination-cue intensity amplified.

Table A3: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. "Pred. join (surv.)" is the classifier's predicted join rate when applied to surveillance-treatment briefings; "Actual" is the LLM's observed rate. The gap measures the surveillance wedge invisible to text classifiers.

Classifier	Acc.	AUC	Pred. join (surv.)	Actual (surv.)	Gap
BoW TF-IDF	88.4%	0.947	40.5%	27.6%	13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	-19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Gap = classifier-predicted join rate − actual LLM join rate under surveillance (pp).

F.3 Information Design with Communication

In a communication version of the information-design grid, upper censorship yields 15.1% joining and lower censorship 17.7% (baseline 3.0%), consistent with censorship making the social-information channel relatively more important.

F.4 Group-Size Awareness

With group-size knowledge ("You are one of 25 citizens"), the pure join rate rises to 0.507 (vs. 0.369 baseline) and the communication premium reverses (−3.4 pp). Monotone signal response is preserved.

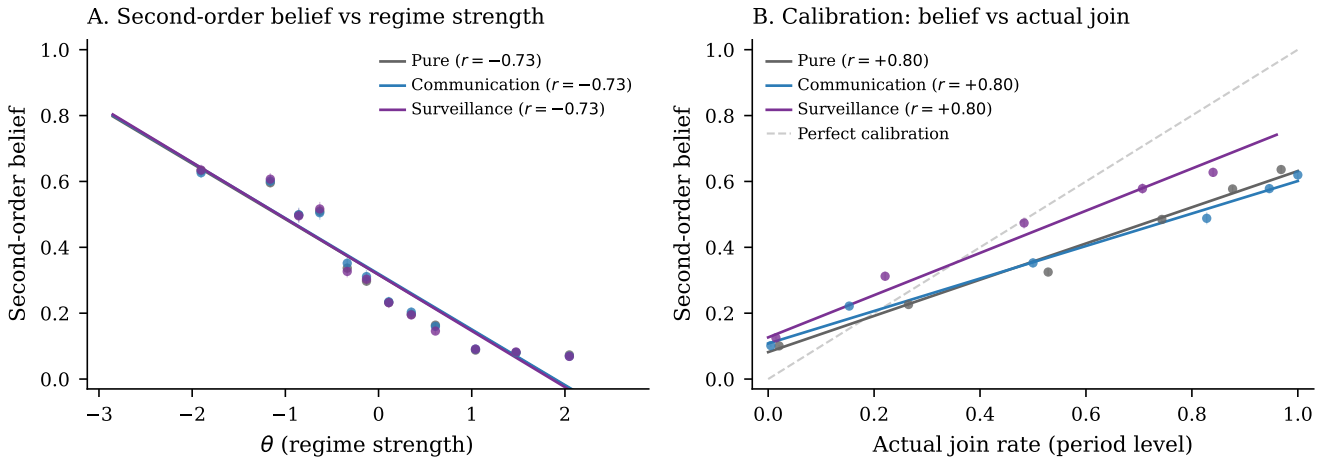


Figure A2: Second-order beliefs. Surveillance overlaps with pure; no significant shift.

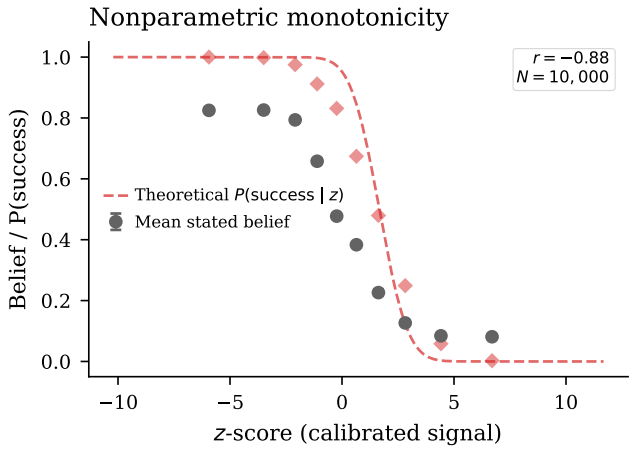


Figure A3: Nonparametric monotonicity: mean belief by z-score bin vs. theoretical benchmark.

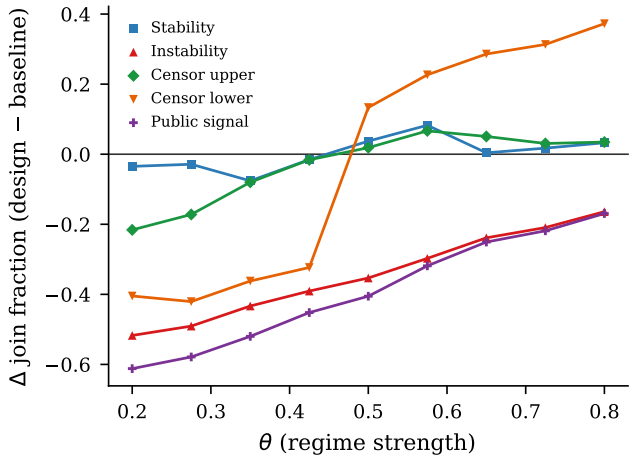


Figure A4: Treatment effect $\Delta(\theta)$ by information design.

Table A4: B/C comparative statics: classifier vs. actual LLM behavior. A logistic trained on baseline join rates (which captures the same information as slider features) predicts similar join rates across all payoff conditions. Actual LLM behavior shifts by ≈ 50 pp, demonstrating that agents respond to payoff information not captured by text features.

Condition	N	Classifier pred.	Actual	Gap (pp)
Baseline ($\theta^* = 0.50$)	270	40.9%	40.9%	+0.0
High cost ($\theta^* = 0.25$)	270	40.9%	19.0%	+21.9
Low cost ($\theta^* = 0.75$)	270	40.9%	69.3%	-28.4

Notes: Primary model (Mistral Small Creative). Logistic classifier trained on baseline slider features. Gap = classifier-predicted join rate − actual LLM join rate (pp).

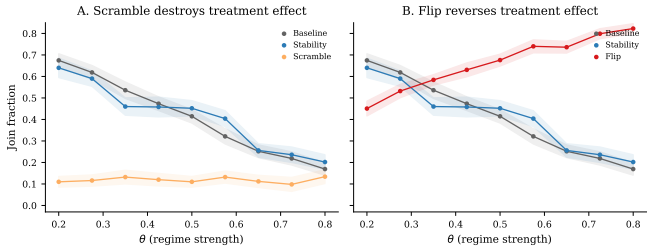


Figure A5: Falsification within information design: scramble ($r = -0.033$), flip ($r = -0.820$).

F.5 Censorship with Common Knowledge

Making censorship common knowledge ("regime censors are suppressing unfavorable intelligence") nearly eliminates the pooling effect: mean join rate returns to 43.2% (+2.4 pp vs. baseline) compared with 37.7% (−3.1 pp) under naïve censorship (Table A11). The behavioral shift requires agents to be naïve about the censorship rule.

The correlation $r(J, A(\theta))$ ranges from +0.76 to +0.84 across 13 model–temperature combinations ($T \in$

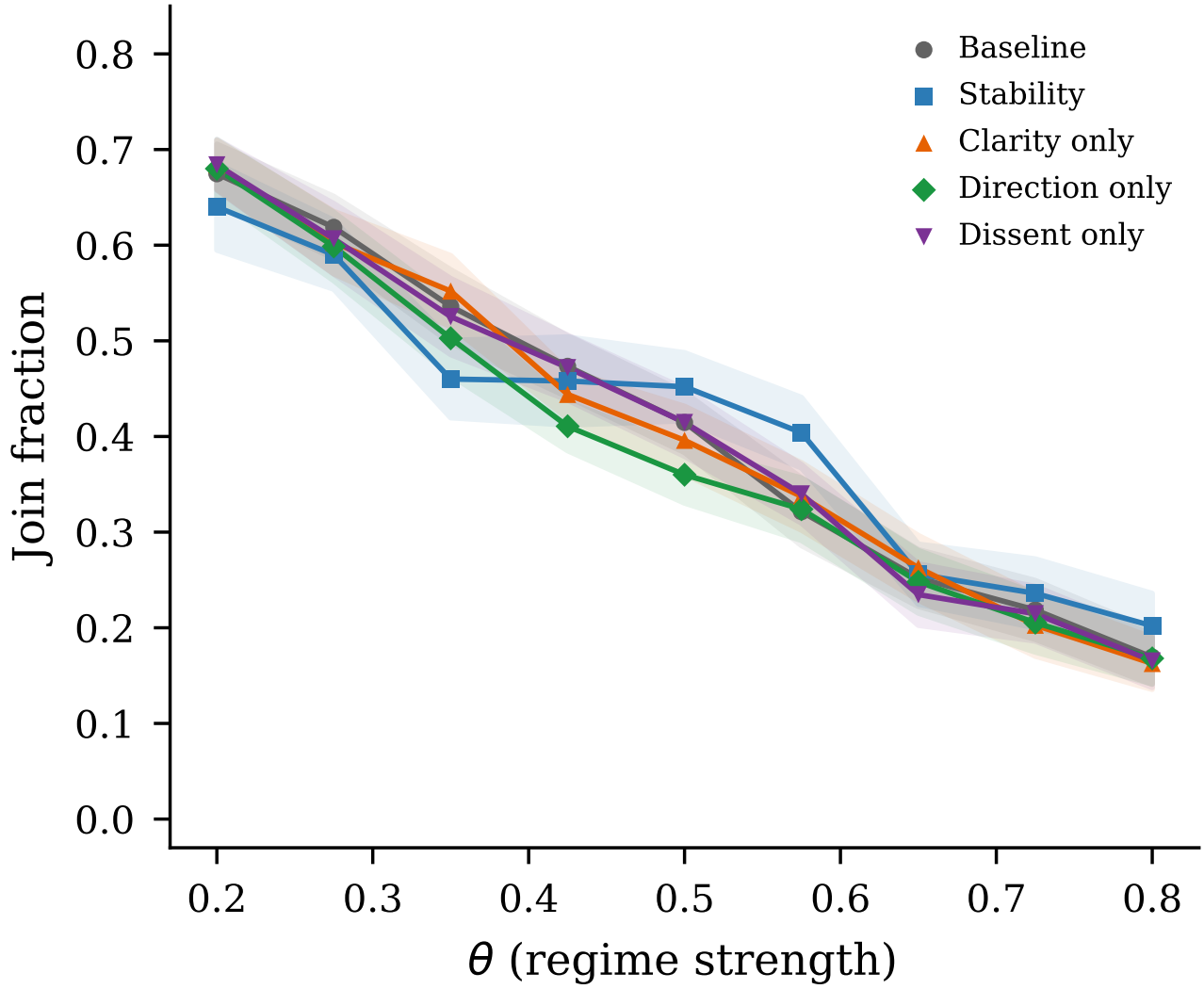


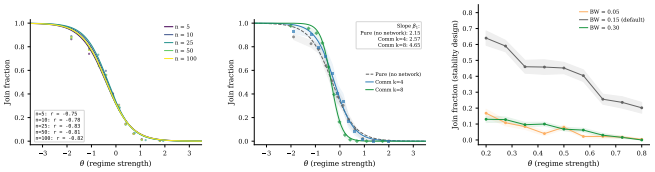
Figure A6: Single-channel decomposition: individual effects are modest (-0.2 to -2.0 pp); the combined effect (-9.0 pp) shows strong super-additivity.

Table A5: Single-channel decomposition of the stability design (primary model: Mistral Small Creative).

Channel	Mean	r	Δ (pp)
Full stability	0.319	+0.62	-9.0
Clarity only	0.405	+0.89	-0.4
Direction only	0.389	+0.87	-2.0
Dissent only	0.406	+0.88	-0.2
Sum of channels	—	—	-2.6
Full design	—	—	-9.0

Notes: Each row is a separate infodesign run for Mistral Small Creative on the same θ grid as Table 6. Δ reports the mean difference vs. the baseline infodesign mean (Table 6).

$\{0.3, 0.5, 0.7, 1.0, 1.2\}$ for three models; Table A12).



(a) Agent count. (b) Network density. (c) Bandwidth.

Figure A7: Robustness checks for threshold-policy alignment and treatment effects.

F.6 Falsification and Placebo Tests

Two placebo variants—monitoring "for research purposes" (no consequences) and "aggregated anonymously"—produce no significant deviation from the communication baseline (+2.6 pp, $p = 0.4164$ and +4.0 pp, $p = 0.2006$; Ta-

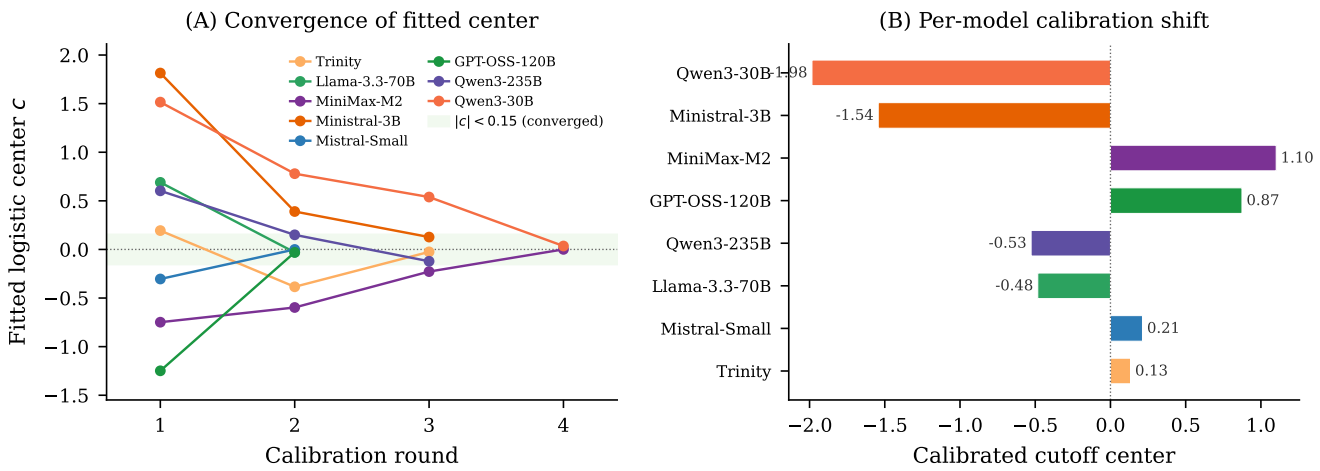


Figure A8: Calibration convergence. *Left:* Trajectory of fitted logistic center across autocalibration rounds. *Right:* Final calibrated cutoff center per model.

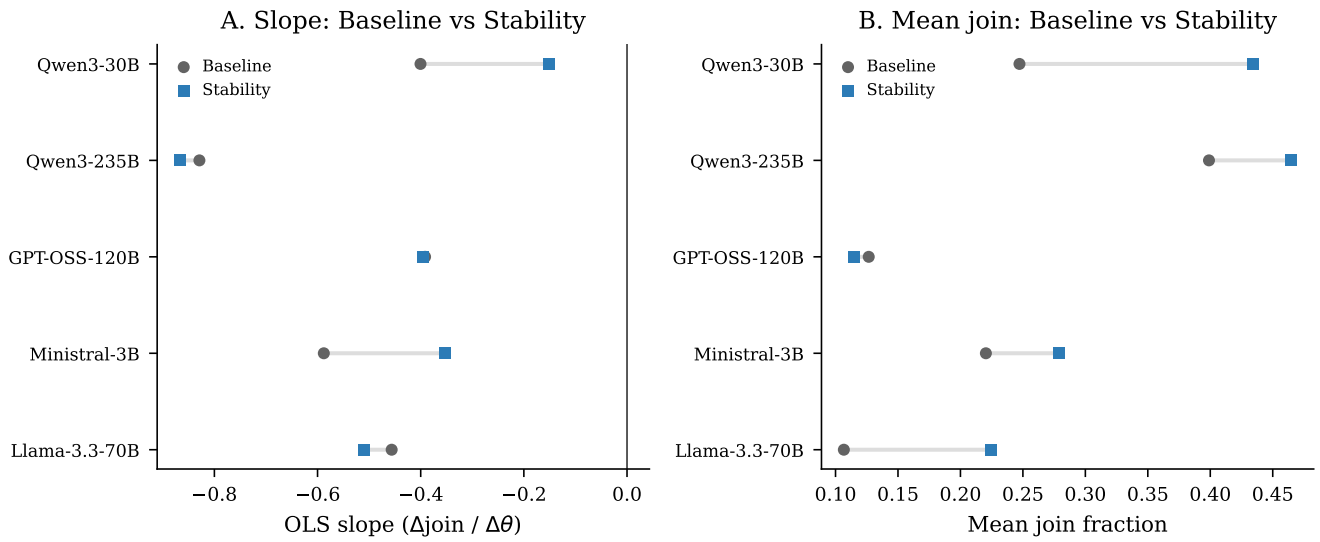


Figure A9: Cross-model replication of information design treatments.

ble A13). The chilling effect is specific to the surveillance warning's strategic content.

Three tests confirm the signal is distributed across all eight evidence domains. Observation shuffle ($r = +0.912$), domain scramble for coordination bullets ($r = +0.921$), and domain scramble for state-capacity bullets ($r = +0.928$) all preserve the baseline correlation ($r = +0.885$), ruling out bullet ordering or any single domain subset as the driver.

F.7 Finite-N Benchmark

Predicted regime fall rates (binomial model with out-of-sample logistic fits) correlate with empirical rates at $r \geq 0.88$ for every model (Mistral: $r = 0.9995$; Table A14).

F.8 Agent-Level Analysis

Table A15 reports agent-level logit regressions (N = 287,055, clustered SEs). All treatment effects are significant and correctly signed. Elicited beliefs predict actions beyond what the signal alone predicts (Column 3).

Figure A10(A) shows a three-feature model (direction, clarity, coordination) outperforms a one-feature sentiment baseline; Panel (B) confirms cross-treatment generalization from pure to communication and surveillance.

F.9 Cross-Generator Robustness

Three text rendering formats—baseline narrative, diplomatic cable, journalistic wire—use identical slider functions but differ in surface prose. Correlations are virtually identical (max within-model difference 0.02; Figure A11,

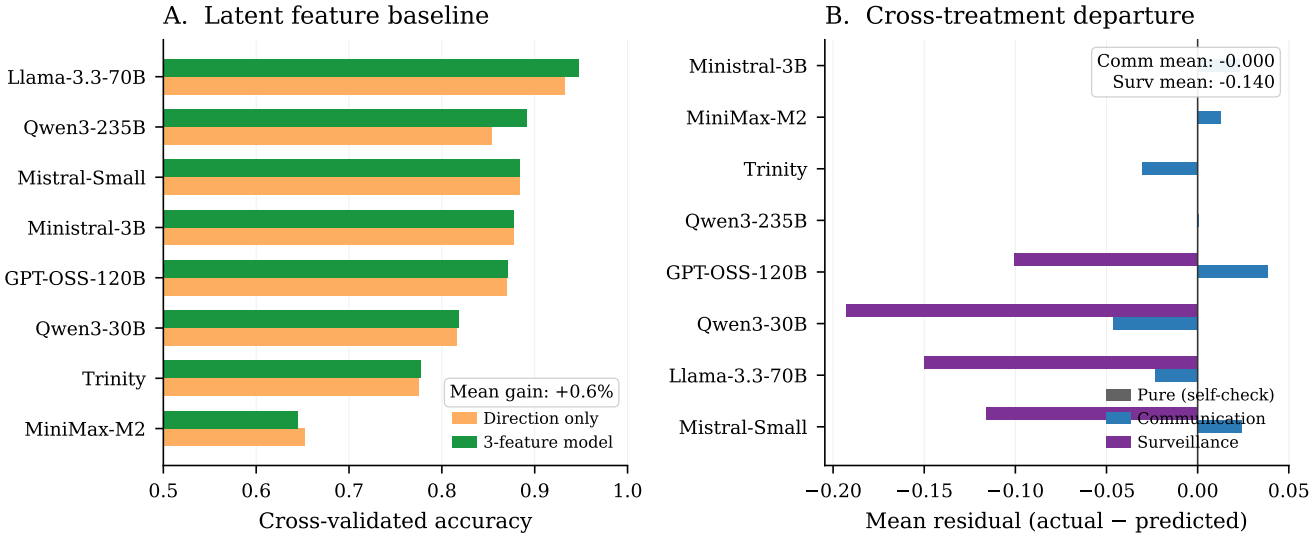


Figure A10: Construct validity tests. (A) Three-feature vs. one-feature prediction accuracy. (B) Cross-treatment generalization.

Table A6: Uncalibrated robustness: models run without any calibration adjustment. Even without calibration, six of seven models show strong $r(J, A(\theta))$, confirming that the sigmoid is not an artifact of the calibration procedure.

Model	N	Mean join	$r(J, A(\theta))$	p
Mistral Small Creative	100	0.382	+0.81	0.0000
Llama 3.3 70B	100	0.422	+0.83	0.0000
Qwen3 30B	100	0.585	+0.90	0.0000
GPT-OSS 120B	100	0.336	+0.77	0.0000
Qwen3 235B	100	0.445	+0.86	0.0000
MiniMax M2-Her	100	0.425	+0.71	0.0000

Notes: Pure treatment, no calibration adjustment (cutoff center = 0). Trinity Large excluded due to elevated API error rates. $r = r(J, A(\theta))$.

Table A7: Placebo calibration. The cutoff center is deliberately shifted by ±0.3 from its calibrated value. The action–attack-mass correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts, confirming that calibration does not create the sigmoid.

Model	Condition	N	Mean join	$r(J, A(\theta))$
Mistral Small Creative	Calibrated	—	0.387	+0.75
	$\Delta c = +0.3$	100	0.424	+0.76
	$\Delta c = -0.3$	100	0.362	+0.76
Llama 3.3 70B	Calibrated	—	0.439	+0.82
	$\Delta c = +0.3$	100	0.385	+0.75
	$\Delta c = -0.3$	100	0.340	+0.76

Notes: Pure treatment. $\Delta c = \pm 0.3$: deliberate shift from calibrated cutoff center. Correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts.

Table A17), ruling out prose style as the driver.

F.10 Belief and Risk Elicitation

Stated beliefs track the theoretical success probability ($r_{\text{post}} = +0.78$) and predict actions ($r_{\text{b,d}} = +0.84$; partial $r = +0.64$ controlling for signal). Under surveillance, beliefs shift only −0.7 pp while actions shift −12.9 pp (Table A18). Pre-decision elicitation ($N = 5{,}000$) produces $r_{\text{pre}} = +0.82$ vs. $r_{\text{post}} = +0.84$, ruling out ex-post rationalization.

Mean punishment risk is ≈8.0/10 across all conditions with negligible JOIN/STAY differences (< 0.2 points; Table A19), supporting the contaminated-communication interpretation.

F.11 Briefing Generator Examples

Table A20 reports the three slider values (direction, clarity, coordination) at representative z-scores. The direction slider is logistic in z (slope 0.8, centered at 0). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the cutoff (maximally ambiguous) and approaches 1 in the extremes (maximally clear). Coordination is logistic in z (slope 0.6, centered at 0) and decreases in z: weak-regime signals imply a more open coordination climate. All three jointly determine phrase selection across eight evidence domains.

At the extremes ($|z| = 2$), clarity is high (0.98) and evidence is largely one-sided: $z = -2$ selects phrases emphasizing regime fragility ("visible cracks in the inner circle") and collective readiness, while $z = +2$ selects regime-strength phrases ("inner circle appears unified") and collective caution. At $z = 0$, clarity is 0.00 (maximally

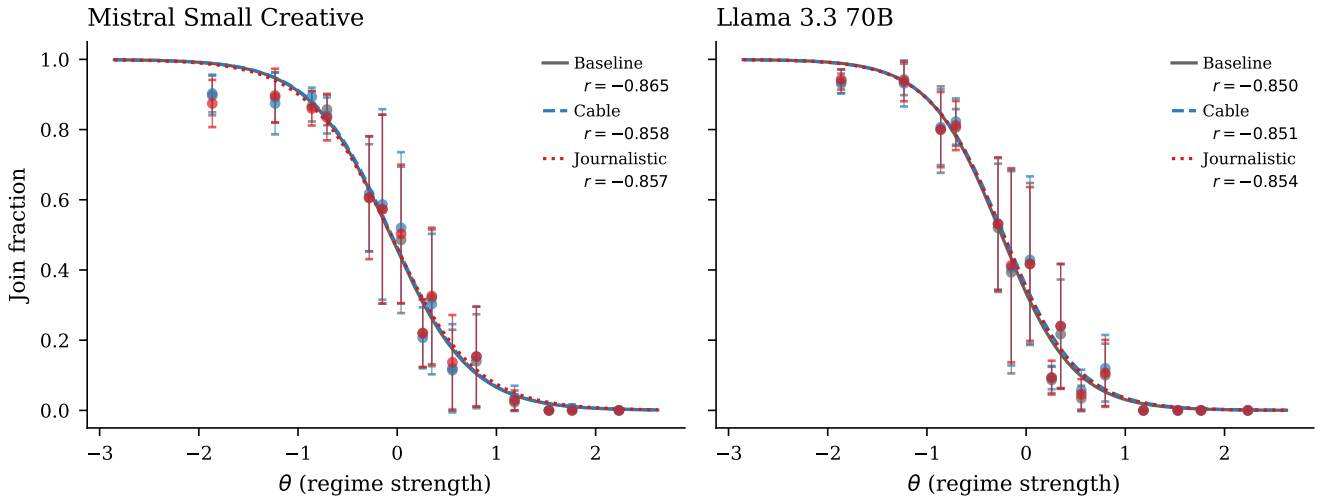


Figure A11: Cross-generator robustness. Three text formats produce indistinguishable sigmoids.

Table A8: Calibration robustness. r_θ: raw correlation with regime strength. r_A: correlation with theoretical attack mass. RMSE: root mean squared error vs. $A(\theta)$. Text slope: logistic slope of naïve $1 - \text{direction}$ predictor.

Model	r_θ	r_A	RMSE	Text slope
Mistral Small Creative	−0.81	+0.67	0.354	-9.2
Llama 3.3 70B	−0.85	+0.79	0.288	-25.1
Qwen3 30B	−0.84	+0.78	0.287	-7.5
GPT-OSS 120B	−0.84	+0.70	0.359	-8.2
Qwen3 235B	−0.85	+0.70	0.354	-20.5
Trinity Large	−0.87	+0.84	0.262	-4.0
MiniMax M2-Her	−0.79	+0.66	0.360	-2.1

Notes: r_θ: Pearson correlation with regime strength θ. r_A: correlation with theoretical attack mass $A(\theta)$. Text slope: logistic slope of naïve $1 - \text{direction}$ predictor (the direction slider inverted as a baseline text classifier). All models use pure treatment with calibrated parameters.

Table A9: Bandwidth robustness: treatment effects Δ (pp, treatment − baseline) within each bandwidth condition (primary model: Mistral Small Creative). Top row shows baseline join rates for reference.

	BW=0.05	BW=0.15	BW=0.30
Baseline (level)	0.054	0.409	0.061
Treatment effect Δ *(treatment − baseline):*			
Stability	+0.7	-9.0	+0.9
Upper cens.	+6.2	-3.1	+5.3
Lower cens.	+10.1	-1.9	+9.6

Notes: Primary model, information design θ-grid $[0.20, 0.80]$. h = Gaussian proximity bandwidth controlling how broadly manipulation spreads around θ^*. Δ: change in mean join vs. bandwidth-specific baseline (pp).

ambiguous) and the briefing presents a balanced picture.

G Implementation Details

All LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model versions are those available December 2024–February 2025.

The *decision prompt* instructs: "You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning." The *communication prompt* asks agents to "write a brief message (1–2 sentences) to share your read of the situation" with trusted contacts. The *surveillance* variant appends: "You have reason to believe that your communications are being monitored by regime security services. Messages deemed subversive could have serious consequences." The *cost/benefit* test prepends a high-cost header ("severe reprisals—imprisonment, asset seizure") or low-cost header ("minimal consequences—brief detentions at most") to the standard briefing.

Each country has a base prior $\bar{z} \sim \mathcal{N}(0, 0.3)$; each period perturbs it by $\mathcal{N}(0, 0.05^2)$. Regime strength is $\theta \sim \mathcal{N}(\bar{z}, 1)$; private signals are $x_i = \theta + \varepsilon_i$ with $\sigma = 0.3$. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), regenerated each period. All random draws use a master seed logged in per-run JSON manifests; LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% for five of seven models; Trinity Large has elevated API errors ($\approx 9\%$) due to content filtering. All statistics use `join_fraction_valid` (Table A21).

All code, prompts, cached responses, and output data are available at https://github.com/keltokhy/llm-global-games.

Table A10: Cross-model replication of key information design conditions. r is the correlation $r(J, A(\theta))$ between join fraction and theoretical attack mass.

	Baseline		Scramble		Flip	
Model	Mean	r	Mean	r	Mean	r
Mistral Small Creative	0.409	+0.88	0.121	−0.03	0.663	−0.82
GPT-OSS 120B	0.127	+0.80	0.132	−0.08	0.677	−0.75
Llama 3.3 70B	0.107	+0.81	0.105	+0.81	0.887	−0.72
Qwen3 30B	0.247	+0.61	0.279	+0.12	0.784	−0.85
Qwen3 235B	0.399	+0.88	0.394	+0.02	0.430	−0.87

Notes: Information design θ-grid $[0.20, 0.80]$. $r = r(J, A(\theta))$: Pearson correlation between join fraction and theoretical attack mass. $N = 25$ agents per period.

Table A11: Censorship with and without common knowledge. Naïve: agents do not know censorship is active. Known: agents are told that regime censors suppress unfavorable intelligence above a severity threshold.

Design	N	Mean join	$r(J, A(\theta))$	Δ (pp)
Baseline (no censorship)	270	0.409	+0.88	—
Upper censorship (naïve)	270	0.377	+0.72	-3.1
Upper censorship (known)	270	0.432	+0.74	+2.4

Notes: Primary model, information design θ-grid. "Known": agents are told that the regime censors intelligence above a severity threshold. Δ: change vs. baseline (pp).

Table A12: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small	T=0.3	100	0.412	+0.78	-0.049
Mistral Small	T=0.7	100	0.406	+0.76	-0.079
Mistral Small	T=1.0	100	0.410	+0.79	-0.039
Llama 70B	T=0.3	100	0.364	+0.78	-0.210
Llama 70B	T=0.5	100	0.360	+0.78	-0.219
Llama 70B	T=0.7	100	0.361	+0.78	-0.220
Llama 70B	T=1.0	100	0.360	+0.78	-0.215
Llama 70B	T=1.2	100	0.358	+0.78	-0.227
Qwen 235B	T=0.3	100	0.392	+0.83	-0.117
Qwen 235B	T=0.5	100	0.392	+0.82	-0.112
Qwen 235B	T=0.7	100	0.392	+0.84	-0.110
Qwen 235B	T=1.0	100	0.398	+0.83	-0.105
Qwen 235B	T=1.2	100	0.397	+0.83	-0.121

Notes: Pure treatment, calibrated parameters, varying LLM decoding temperature across three models.

H Implications for AI Alignment

Three patterns carry implications for AI safety. First, surveillance induces strategic self-censorship (belief–action wedge) without any training signal for deceptive behavior, suggesting that such capabilities emerge from pretraining on human strategic text rather than explicit optimization. Second, stated beliefs mediate behavior (Pseudo $R^2 = 0.975$), so alignment techniques monitoring only inputs and outputs may miss decision-relevant computation.

Table A13: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Variant	N	Mean join	$r(J, A(\theta))$	Δ (pp)	p
Placebo	200	0.437	+0.75	+2.6	0.416
Anonymous	200	0.452	+0.77	+4.0	0.201

Notes: Primary model (Mistral Small Creative). Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. Δ: change vs. communication baseline (pp). p-value from two-sample t-test.

Table A14: Finite-N Benchmark: Predicted vs. Empirical Regime Fall Rates

Model	N periods	Logistic x_0	Pearson r	RMSE	MAE
Mistral	599	−0.11	0.997^{***}	0.042	0.015
Llama 70B	99	0.05	0.954^{***}	0.156	0.059
Qwen 30B	99	0.20	0.986^{***}	0.088	0.029
GPT-OSS 120B	199	−0.01	0.997^{***}	0.038	0.014
Qwen 235B	199	−0.03	0.989^{***}	0.073	0.025
Trinity	99	0.19	0.979^{***}	0.108	0.043
MiniMax	99	1.02	0.999^{***}	0.028	0.013
Pooled	1399	−0.05	0.999^{***}	0.019	0.007

Notes: For each θ bin, the predicted fall rate is $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. Pearson r measures correlation between predicted and empirical fall rates across θ bins. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Third, information-structural manipulation (censorship, ambiguity, public signals) shifts behavior by up to 40 pp, implying that adversarial prompt design can exploit the same channels as alignment interventions. Whether the belief–action wedge constitutes genuine preference falsification (Kuran, 1991) or pattern-matching remains open; the relevant finding is its *robustness* across architectures.

Table A15: Agent-Level Regressions

	(1) Join Decision Logit		(2) Coordination Logit		(3) Belief → Action Logit	
θ	-1.742^{***}	(0.043)				
Direction			-14.088^{***}	(0.996)		
Coordination			-9.333^{***}	(1.131)		
Dir × Coord			-0.042	(0.360)		
Belief					0.367^{***}	(0.021)
z-score					-0.040	(0.077)
Comm	0.024	(0.023)				
Flip	0.008	(0.056)				
Propaganda K10	-0.381^{***}	(0.069)				
Propaganda K2	-0.230^{***}	(0.073)				
Propaganda K5	-1.049^{***}	(0.068)				
Propaganda Surveillance	-1.544^{***}	(0.062)				
Scramble	0.065	(0.040)				
Surveillance	-1.546^{***}	(0.048)				
$\theta\times$ Comm	-0.465^{***}	(0.041)				
$\theta\times$ Flip	3.305^{***}	(0.071)				
$\theta\times$ Propaganda K10	-1.022^{***}	(0.107)				
$\theta\times$ Propaganda K2	-1.115^{***}	(0.113)				
$\theta\times$ Propaganda K5	-1.941^{***}	(0.092)				
$\theta\times$ Propaganda Surveillance	-0.260^{***}	(0.082)				
$\theta\times$ Scramble	1.698^{***}	(0.044)				
$\theta\times$ Surveillance	-1.153^{***}	(0.062)				
Beliefs Propaganda K5 (belief)					0.009	(0.244)
Beliefs Pure (belief)					4.819^{***}	(1.106)
Beliefs Surveillance (belief)					0.028	(0.253)
Constant	-0.090^{*}	(0.051)	11.022^{***}	(1.065)	-26.495^{***}	(1.581)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	287,055		44,662		18,990	
Pseudo R^2	0.384		0.442		0.975	

Notes: Logit coefficients reported with clustered standard errors (model–country–period) in parentheses. Column (1): agent-level join decision on θ, treatment dummies, and interactions, with model fixed effects. Base category: pure treatment. Column (2): coordination ablation using briefing slider values (pure treatment only). Column (3): partial effect of elicited belief on action, controlling for z-score. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A16: Logistic fit parameters by model and treatment. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); β is the logistic slope. Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

	Pure		Communication	
Model	$\hat{\theta}^*$ (SE)	β (SE)	$\hat{\theta}^*$ (SE)	β (SE)
Mistral Small Creative	-0.32 (0.02)	$+2.15$ (0.08)	-0.23 (0.02)	$+2.57$ (0.11)
Llama 3.3 70B	$+0.02$ (0.04)	$+2.83$ (0.36)	-0.06 (0.04)	$+3.63$ (0.52)
Qwen3 30B	$+0.10$ (0.06)	$+1.62$ (0.16)	-0.03 (0.04)	$+2.94$ (0.36)
GPT-OSS 120B	-0.25 (0.04)	$+2.06$ (0.15)	-0.15 (0.03)	$+3.03$ (0.26)
Qwen3 235B	-0.22 (0.03)	$+2.11$ (0.15)	-0.22 (0.03)	$+2.62$ (0.23)
Trinity Large	$+0.08$ (0.05)	$+1.34$ (0.11)	-0.02 (0.04)	$+2.23$ (0.23)
MiniMax M2-Her	-0.17 (0.09)	$+0.61$ (0.06)	-0.07 (0.09)	$+0.66$ (0.06)

Notes: $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\beta \equiv b_1$: logistic slope coefficient. Standard errors from the covariance matrix of the logistic fit; cutoff SE by delta method.

Table A17: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small Creative	Baseline	100	0.402	+0.77	-0.070
Mistral Small Creative	Cable	100	0.406	+0.76	-0.057
Mistral Small Creative	Journalistic	100	0.403	+0.75	-0.065
Llama 3.3 70B	Baseline	100	0.354	+0.75	-0.249
Llama 3.3 70B	Cable	100	0.362	+0.75	-0.226
Llama 3.3 70B	Journalistic	100	0.360	+0.76	-0.233

Notes: Information design θ-grid. Three text renderers (baseline intelligence briefing, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs.

Table A18: Agent-level belief correlations (primary model: Mistral Small Creative).

Treatment	N	r_{post}	$r_{\text{b,d}}$	r_{partial}	Mean belief
Pure	10000	+0.78	+0.84	+0.64	0.444
Communication	5000	+0.79	+0.83	+0.56	0.444
Surveillance	5000	+0.78	+0.73	+0.47	0.436

Notes: r_{post}: correlation between posterior and stated belief. $r_{\text{b,d}}$: correlation between belief and decision. r_{partial}: partial correlation of belief and decision controlling for z-score.

Table A19: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. "JOIN" and "STAY" columns show the mean rating conditional on the agent's own decision.

Model	Condition	N	Mean risk	Risk | JOIN	Risk | STAY
Mistral	Pure	2500	8.1	8.0	8.2
Mistral	Comm	2500	8.1	8.0	8.1
Mistral	Surveillance	2500	8.1	8.1	8.1
Llama 70B	Pure	2500	8.0	8.0	7.9
Llama 70B	Comm	2500	7.9	8.0	7.9
Llama 70B	Surveillance	2500	7.9	8.0	7.9

Notes: Agent-level elicitation on a 0–10 scale. Reported risk is the mean rating conditional on the agent's own JOIN or STAY decision.

Table A20: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23

Table A21: Parse error and API failure rates by model and treatment. API error = provider-side failure; unparseable = valid response that could not be classified as JOIN or STAY. Combined rates are below 2% for five of seven models; Trinity Large has elevated API errors (≈9%) due to provider-side content filtering.

Model	Treatment	N	API err	Unparseable
Mistral Small Creative	Pure	1000	0.0%	0.0%
	Comm	1000	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
Llama 3.3 70B	Pure	100	0.0%	0.0%
	Comm	100	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
Qwen3 30B	Pure	100	0.0%	0.0%
	Comm	100	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
GPT-OSS 120B	Pure	200	0.0%	1.5%
	Comm	200	0.0%	0.3%
	Scramble	500	0.0%	3.1%
	Flip	500	0.0%	3.3%
Qwen3 235B	Pure	200	0.0%	0.0%
	Comm	200	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
Trinity Large	Pure	100	9.0%	0.0%
	Comm	100	10.0%	0.0%
	Scramble	100	8.0%	0.0%
	Flip	100	9.9%	0.0%
MiniMax M2-Her	Pure	100	0.0%	1.5%
	Comm	100	0.0%	0.9%
	Scramble	100	0.0%	1.8%
	Flip	100	0.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (timeout, rate limit, content filter). Unparseable = valid LLM response that could not be classified as JOIN or STAY.